NIKHIL REDDY LEVAKU

+91-6303252008 | levakunikhilreddy8@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Software Development Engineer with production experience building **100+ RESTful APIs** using Core Java, Spring Boot, and PostgreSQL across the full **software development lifecycle (SDLC)**. Strong foundation in **data structures & algorithms**, OOP, **microservices architecture**, and event-driven systems (Kafka/RabbitMQ). Hands-on experience in concurrency-safe transactional workflows, database optimization, GitHub Actions CI, and **Agile** delivery.

TECHNICAL SKILLS

Languages & Frameworks	Java, SQL, Spring Boot, Spring Security, Hibernate/JPA
Databases & Cache	PostgreSQL, MySQL, Redis, Flyway
Messaging & Systems	Apache **Kafka**, RabbitMQ, Microservices, Event-Driven Design, RESTful API Design
Testing & DevOps	JUnit 5, Mockito, Postman, Swagger / OpenAPI, Git, GitHub Actions (CI), Jenkins, SonarQube, SonarScanner, Docker, Docker Compose, Kubernetes, Helm 3, Argo CD, Nginx, Terraform (IaC Modules), Ansible, Bash, AWS (EC2, VPC, S3, RDS, CloudWatch), Linux, Gradle, Maven
Concepts	Data Structures & Algorithms, Concurrency, OOP, System Design, SDLC, Agile / Scrum

PROFESSIONAL EXPERIENCE

DeepLure Research Pvt Ltd — Backend Engineer **May 2026 – Aug 2026**

Java · Spring Boot · Spring Security · Microservices · PostgreSQL · Redis · RabbitMQ · Infisical · AWS S3 · Docker

- Analyzed business requirements and shipped **100+ production REST APIs** across B2B and on-demand platforms, building configurable backend components for authentication, orders, payments, and admin modules.
- Implemented **JWT** authentication and **role-based access control (RBAC)**, with OTP verification over SMS and email.
- Integrated the **Razorpay** payment gateway (orders, wallets, refunds), applying idempotency keys and concurrency locking to prevent double-charges and race conditions under concurrent requests.
- Used **RabbitMQ** for asynchronous inter-service notification workflows; managed application secrets with **Infisical**.
- Created **AWS S3** buckets for application media storage.
- Engineered **Redis-backed asynchronous workers** for notifications, payment reconciliation, and booking expiry, and reduced database load **~40%** through Redis caching and query optimization.
- Modeled normalized **PostgreSQL** schemas with Hibernate/JPA and Flyway; aligned API contracts with frontend teams via Swagger/OpenAPI; wrote **JUnit 5** and **Mockito** tests; contributed in Agile sprints.

SmartBridge Educational Services (APSCHE) — Data Analytics Intern **May 2025 – Jul 2025**

- Cleaned sales datasets with SQL joins and aggregates; built 3 Tableau dashboards for production and inventory review.

PROJECTS

Flight Booking System — Distributed Microservices Platform **Aug 2025 – Jan 2026**

Java · Spring Boot · Microservices · PostgreSQL · Redis · Apache Kafka · Docker · Nginx | GitHub

- Refactored a monolithic Spring Boot backend into **4 independently deployable microservices** (Auth, Flight, Booking, Notification), each with its own **PostgreSQL** database and versioned **Flyway** migrations.
- Implemented inter-service REST communication — the Booking Service calls the Flight Service to reserve seats — propagating the user's **JWT** so each service independently authenticates and enforces RBAC (USER/ADMIN).
- Prevented overbooking under concurrent requests with transactional seat reservation using row-level **pessimistic write locking** (**PESSIMISTIC_WRITE**) inside **@Transactional** methods; added seat rollback logic.
- Built an **event-driven** flow with **Apache Kafka**: Booking Service publishes a booking-created event, consumed asynchronously by a dedicated Notification Service, decoupling notifications from the booking transaction.
- Added **Redis caching** in the Flight Service, request validation, centralized exception handling, and **JUnit 5/Mockito** tests.
- Containerized each service with multi-stage **Docker** builds, configured the EC2 server with **Ansible**, and ran the stack with **Docker Compose** behind an **Nginx** reverse proxy; added a **GitHub Actions** CI workflow.

User Management System **Oct 2024 – Dec 2024**

Spring Boot · Hibernate/JPA · MySQL | GitHub

- Built a CRUD REST API with dynamic search and filtering, request validation, and centralized exception handling.
- Modeled a normalized 6-table MySQL schema with JPA entity mappings; tested and documented all endpoints in Postman.

EDUCATION

Sri Venkateswara College of Engineering, Tirupati **2022 – May 2026**

B.Tech, Electronics & Communication Engineering CGPA: 8.5 / 10

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate — Oracle (Sep 2025) · Problem Solving, Intermediate — HackerRank (Apr 2026) · SQL, Intermediate — HackerRank (Mar 2026) · Java Programming — GeeksforGeeks (Mar 2026)

ACHIEVEMENTS

- Solved **100+ Data Structures & Algorithms** problems on LeetCode; cleared HackerRank Problem Solving (Intermediate).
- Built a Spring Boot LMS backend at **Design Spark Challenge 2026**, a 48-hour innovation marathon (RTIH, Andhra Pradesh).